

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science and Engineering
MST Exam-March'26

BE: Semester (II)

Course Code: UAI201; Course Name: Programming Essentials for AI & ML

14th March'2026(3:00 PM)

Faculty: Dr. Palika Chopra(X21-X24), Dr. Ashima Khosla(X11-X14)

Time: 2 hours

Marks: 5*6=30

Note: Attempt any five questions. <u>Solve in order</u> , new problem, new page, if not sure, leave appropriate pages in between. Plan on the last page to avoid cutting.			CO, BL
Q1:	a) Write a Python program to find the factorial of a given number using a for loop. If the user enters 5, the output should be: Factorial = 120 b) Write a Python program to check whether a number is a Neon number. Example: 9 9 ² =81 8 + 1 = 9	3,3	CO2, BL2
Q2:	a) Write a Python program to copy the entire contents of one text file (source.txt) into another text file (destination.txt). Use appropriate file opening modes and ensure proper file handling. b) Write a Python program to read a text file named data.txt and print only those lines that contain the word "Python". Also display the file pointer position before reading each line using the tell() function. Example: If the file data.txt contains: I like programming Python is easy to learn Java is my favourite too Python is widely used in AI The output should display only the lines containing "Python" along with their file pointer positions of that line.	3,3	CO3, BL4
Q3:	Distinguish between the following pairs by stating any two valid differences in each case. Use syntax and examples wherever applicable. i) for and while loops ii) break and continue statements iii) Compiler and Interpreter	2*3=6	CO2, BL2
Q4:	a) Write a Python program that asks the user to enter two numbers and performs division. Handle the ZeroDivisionError exception and display an appropriate message if the user attempts to divide a number by zero. b) Write a Python program to create a user-defined exception named InvalidMarksError. Raise this exception if the marks entered by the user are less than 0 or greater than 100. Display an appropriate error message.	3,3	CO3, BL4
Q5:	Differentiate between the following pairs by writing any two valid differences for each: a) Compile time errors and Run time errors b) Pickling and Unpickling c) Mutable and Immutable objects	2*3=6	CO1, BL3
Q6:	a) Write a Python program to enter the names of employees and their salaries from the user and store them in a dictionary. Display the dictionary at the end. b) Write a Python program to create a tuple of five numbers from user and display the largest number from the tuple. c) Write a Python program to enter a string and count the number of vowels in it.	2*3=6	CO1, BL3

*****All the Best*****